

Compiler Construction

Lasse Arpe Kristensen (lakri18)

Carl-Emil Dons Christensen (carlc23)

Lucas Bergholdt Hansen (lucha23)

19-12-2025

Contents

1	Introduction	3
2	Lexical Analysis and Scanning	4
3	Syntactic Analysis and Parsing	5
4	Semantic Analysis	6
4.1	Scope Analysis	6
4.2	Type Analysis	6
5	Abstract Syntax Tree and Visitor Pattern	7
6	Interpretation and Tree Traversal	8
7	Error Handling	9
7.1	Lexical Errors	9
7.2	Syntactic Errors	9
7.3	Semantic Errors	9
7.4	Runtime Errors	9
8	Reflection on Potential Optimization	10
9	Conclusion	11

1 Introduction

2 Lexical Analysis and Scanning

3 Syntactic Analysis and Parsing

4 Semantic Analysis

4.1 Scope Analysis

4.2 Type Analysis

5 Abstract Syntax Tree and Visitor Pattern

6 Interpretation and Tree Traversal

7 Error Handling

7.1 Lexical Errors

7.2 Syntactic Errors

7.3 Semantic Errors

7.4 Runtime Errors

8 Reflection on Potential Optimization

9 Conclusion